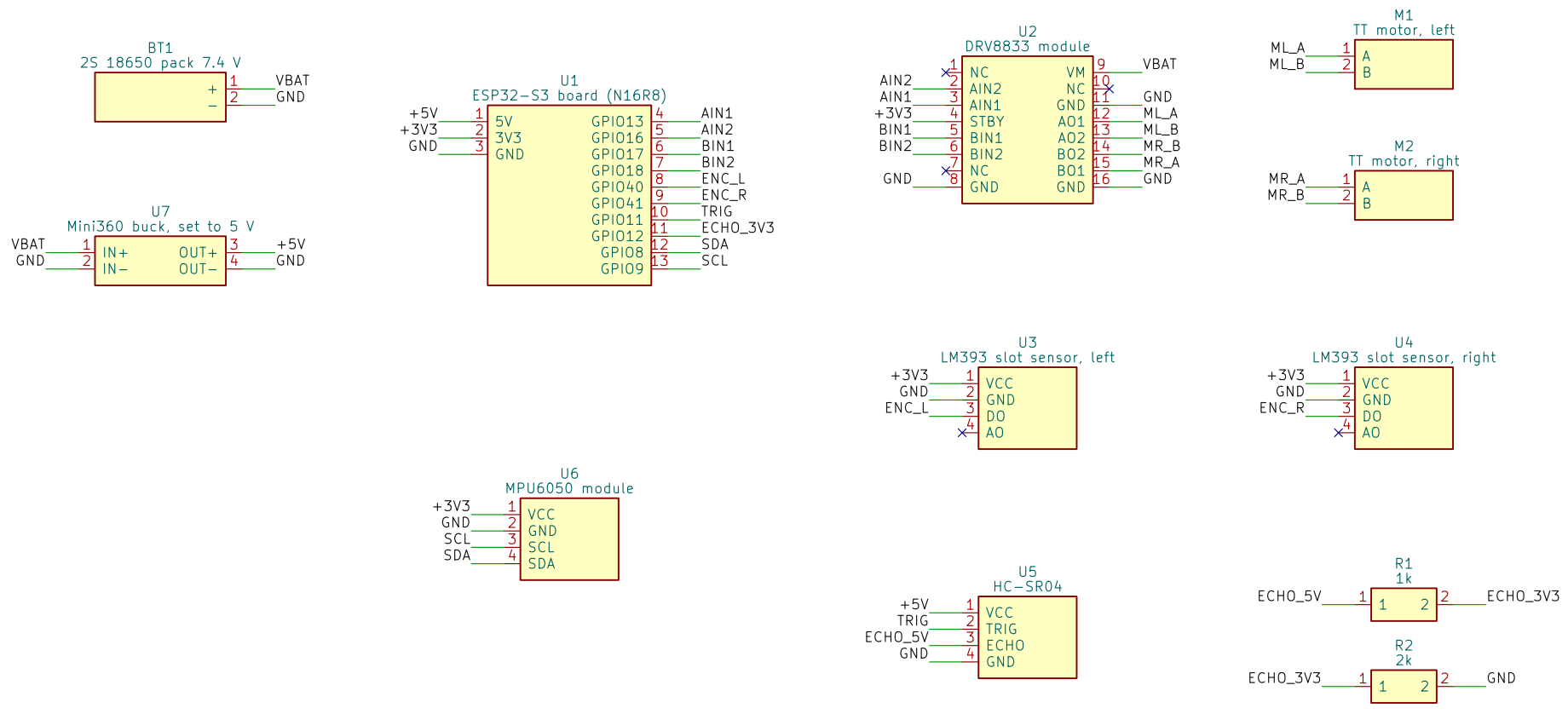
BT1
2S 18650 pack 7.4 V
+
–
1
2
VBAT
GND
U1
ESP32-S3 board (N16R8)
+5V
+3V3
GND
5V
3V3
GND
GPIO13
GPIO16
GPIO17
GPIO18
GPIO40
GPIO41
GPIO11
GPIO12
GPIO8
GPIO9
AIN1
AIN2
BIN1
BIN2
ENC_L
ENC_R
TRIG
ECHO_3V3
SDA
SCL
U2
DRV8833 module
NC
AIN2
AIN1
STBY
BIN1
BIN2
NC
GND
VM
NC
GND
AO1
AO2
BO2
BO1
GND
VBAT
GND
ML_A
ML_B
MR_B
MR_A
GND
M1
TT motor, left
A
B
M2
TT motor, right
U7
Mini360 buck, set to 5 V
IN+
IN–
OUT+
OUT–
VBAT
GND
+5V
GND
U3
LM393 slot sensor, left
VCC
GND
DO
AO
+3V3
GND
ENC_L
U4
LM393 slot sensor, right
ENC_R
U6
MPU6050 module
SCL
SDA
U5
HC-SR04
TRIG
ECHO_5V
ECHO
R1
1k
ECHO_3V3
R2
2k